

SAHIL RAOSAHEB SONAWANE

AI/ML & Generative AI Developer | Computer Engineering Student

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PROFESSIONAL SUMMARY

Third-year Computer Engineering student working in machine learning, generative AI and computer vision. Built an LLM-based learning assistant on the Google Gemini API, a FastAPI decision-support system, and real-time pose and hand-tracking projects with OpenCV and MediaPipe. Comfortable building the backend and React frontend around them.

EDUCATION

B.E. Computer Engineering

Third Year (T.E.)

KJEL's Trinity College of Engineering and Research, Pune

TECHNICAL SKILLS

AI / ML: Artificial Intelligence, Machine Learning, Generative AI, LLM Applications, Google Gemini API, AI Assistants

Computer Vision: OpenCV, MediaPipe, Image Processing, Pose Estimation, Gesture Recognition

Programming: Python, JavaScript, TypeScript, Java

Frameworks / Backend: FastAPI, Flask, Node.js, Express.js, React, Vite

Libraries: NumPy, Pandas, Shapely, Pydantic, Pygame, Three.js, Pytest

Databases: MySQL, MongoDB, SQLite, Firebase / Firestore

Tools: Git, GitHub, Docker, VS Code

PROJECTS

MnemOS | AI learning-memory system (hackathon prototype)

github.com/SahilSonawane2605/MnemOS-Hackathon

- Built a prototype that records what a user studies online and lets them query it later. A Chrome extension (Manifest V3) captures pages, and a Node.js/Express API stores them in SQLite.
 - Integrated Gemini 2.5 Flash as an assistant that answers questions like "What have I been learning recently?" from the stored history, and generates skill trends and career suggestions.
 - Created a React 19 and TypeScript dashboard with a learning timeline and charts. Deployed on Vercel and Render.
- Tech: Google Gemini API, Node.js, Express.js, SQLite, React, TypeScript, Vite, Chrome Extension

SURAKSHA-SETU | Disaster management decision-support system

github.com/SahilSonawane2605/Suraksha-Setu

- Built a FastAPI backend that scores habitation-level flood and landslide risk from 0 to 100 and assigns a risk level and mitigation priority using deterministic rules.
 - Implemented hazard-zone analysis on GeoJSON polygons with Shapely: point-in-polygon checks, intersection tests and Haversine distance calculations.
 - Designed a five-stage relocation engine that screens shelters, checks capacity against the affected population and returns an evacuation recommendation.
 - Added Pandas/NumPy ingestion, Pydantic validation and Pytest tests for the risk, capacity, GIS and API layers. Runs on synthetic demo data; a React frontend consumes the APIs.
- Tech: Python, FastAPI, React, SQLite, Shapely, Pandas, NumPy, Pytest. Smart India Hackathon problem statement SIH26191.

OpenCV Boxing Game (ShadowStrike Ultra) | Motion-controlled game

github.com/SahilSonawane2605/opencv-boxing-game

- Built a webcam boxing game controlled by body movement. MediaPipe Pose tracks both hands, and punches are detected from hand speed and checked for collisions. Pose estimation runs on its own thread.
 - Added enemies with HP and wave progression, combo multipliers, scoring, and particle, screen-shake and slow-motion effects. Sound effects are generated in code with Pygame.
- Tech: Python, OpenCV, MediaPipe, NumPy, Pygame

GestureVerse | Gesture-controlled particle worlds

github.com/SahilSonawane2605/GestureVerse

- Built a browser experience controlled by hand gestures. MediaPipe hand tracking reads landmarks from the webcam, and recognized gestures switch between six particle-based scenes.
 - Rendered a system of about 38,000 particles in Three.js with custom WebGL shaders and bloom post-processing.
- Tech: JavaScript, Three.js, WebGL, MediaPipe, HTML/CSS

Dr. Strange Magical Circle | Real-time hand-tracking filter

github.com/SahilSonawane2605/Dr.Strange_magical_circle

- Created a webcam filter that finds hand landmarks with MediaPipe and triggers the effect when finger positions and distance ratios match a target hand pose.
 - Overlaid two transparent PNG circles, rotating in opposite directions, that follow the hand each frame.
- Tech: Python, OpenCV, MediaPipe, NumPy

RESEARCH / TECHNICAL WORK

Agentic Learning Digital Twins: A Framework for Intelligent Personalized Learning

Technical Seminar

- Seminar work on using agentic AI and digital-twin concepts to personalize learning.
- Used MnemOS as the working prototype for selected ideas, such as learning-activity capture and LLM-based analysis.